Xiangdong (Peter) Xie

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PhD student in Computer Science with research experience in machine learning, computer vision, and multimodal sensing. Built end-to-end pipelines for data collection, preprocessing, statistical analysis, and model development across industrial and health-related applications. Interested in robust, interpretable ML systems for real-world sensing problems.

Selected Technical Skills

- **Programming:** Python, C/C++, Java,
- **ML / CV:** PyTorch, scikit-learn, OpenCV, YOLO, NumPy, pandas
- **Data & Research Tools:** Git, LaTeX, matplotlib
- **Methodology:** signal processing, anomaly detection, object detection, segmentation, wearable sensing, statistical analysis, Bayesian variational inference.

Work/Lab Experience

Intelligent Sensing Lab, UMass Amherst **Amherst, MA**

Research Assistant Feb 2024 – Present

- Co-led/contributed to ML/DL research projects in industrial and medical sensing, including micro-contact printing metrology, submerged pump fault detection, and wearable pain assessment for Alzheimer's Disease and Related Dementia (ADRD) populations.
- Developed visualization, registration, and statistical width-estimation pipelines for condensation-figure metrology, achieving up to 93% line-width accuracy on real data.
- Designed statistical analyses to evaluate the benefit of impeller-mounted sensors and the effect of encapsulation on detection performance

ASAP Lab, USC **Los Angeles, CA**

Research Assistant Aug 2023 – Dec 2023

- Conducted meta-learning and outlier detection research: literature synthesis, prototype pipelines, and initial empirical evaluations on benchmark tasks.

Intelligent Sensing Lab, UMass Amherst **Amherst, MA**

Independent Research Investigator Nov 2022 – June 2023

- Proposed a lightweight auto-encoder anomaly detector with pseudo-boosting (feature bagging) for real-time EHMS, improving benchmark performance while minimizing inference latency for IoMT deployment.

Selected Research Projects

- **Pain assessment in ADRD (ongoing):** Led IRB-approved data collection in a seniority facility in Springfield, Massachusetts, built self-report app (iPad/Mac), cleaned and aligned physiological data, and evaluated DL/ML models under novel slackened metrics. Poster presented at the MassAITC Workshop in Frailty: https://massaitc.org/2026/01/14/14-intelligent-non-invasive-pain-assessment-towards-frailty-prevention-in-communication-impaired-populations/

  Dataset Collection: Co-led the collection of a senior-inclusive pain dataset, bridging the gap of missing data from older-adults and people with cognitive impairment in pain physiology studies: 10.17632/m38f8ydn49.2

- **YOLOv8 CF metrology:** Synthetic CF augmentations and RDA improved mAP@75 from 0.211 to 0.875 with accurate width estimates: https://ieeexplore.ieee.org/abstract/document/11197938
- **Submerged Pump Fault Detection:** Designed hypothesis testing and effect-size analyses to quantify diagnostic benefits of impeller-mounted sensors and to demonstrate encapsulation's negligible impact on detection power, achieving average accuracy of 91.3%, AUC-ROC of 0.973 on pump-fault detection experiments:

Coursework

- **Light-weight Hand-pose estimation (class project):** Proposed and implemented an alternative training of the traditional Point Net + LSTM for hand pose classification from 3D-depth images which performs well on scarce dataset, achieving an 87.4% test accuracy on the dataset that is 1% dense as the original DHG dataset, with only 200,000 parameters: https://github.com/straybird16/Hand-Pose-Estimation-Model
- **UI Removement Tools for Child-grooming Detection (class project)**: Implemented the whole dataset generation pipeline for the training of a large-size YOLOv8 model for SnapChat GUI element detection and tuned the training configurations. Added various API functions for the customization of output generation for another diffusion model to in-paint the image with the output: https://youtu.be/GG2-jW0k7Us
- **Relevant coursework:** Advanced Algorithms, Computer Vision, Reinforcement Learning, Machine Learning, Distributed Systems, Quantum Computation: https://arxiv.org/abs/2509.16086

Selected Publications

- Xie, X., Yan, J., Ma, R. and Du, X., 2025. In-Situ Metrology for Roll-to-Roll Microcontact Printing via Condensation Figures and YOLOv8. *IEEE Transactions on Industrial Informatics.*
- Xie, X., Safarzadeh, M., Zhu, M., Riddington, S., Abban, S., Odei-Opoku, H., Adeleye, K., Prasad, S., Chung, J., Du, X., 2026. SILVER-Pain Dataset, *Mendeley Data, V2, 10.17632/m38f8ydn49.2*

Manuscripts in Submission

- Xie, X., Safarzadeh, M., Zhu, M., Riddington, S., Abban, S., Odei-Opoku, H., Adeleye, K., Prasad, S., Chung, J., Du, X., 2026. SILVER (Senior-Inclusive Laboratory Video and Empatica Recordings of) Pain Dataset. Submitted to *Scientific data*
- Wankhede, S.P., Xie, X., Alshehri, A.H., Brashler, K.W., Ba'adani, M., Turcan, D.C., Youcef-Toumi, K. and Du, X., 2025. In-Situ Fault Detection of Submerged Pump Impellers Using Encapsulated Accelerometers and Machine Learning. *arXiv preprint arXiv:2509.16086.* Submitted to *Mechanical Systems and Signal Processing*

Education

University of Massachusetts at Amherst — **Amherst, MA**
Ph.D. in Computer Science — **Expected December 2027**
GPA: 3.7/4.0

University of Massachusetts at Amherst — **Amherst, MA**
Master of Science in Computer Science — **May 2023**
GPA: 3.5/4.0

Grinnell College — **Grinnell, IA**
Bachelor of Arts in Computer Science — May 2021
Major GPA: 3.7/4.0